

1. What are Comments?

Comments are notes written inside a program to explain the code.

Comments are **ignored by the compiler**, so they do not affect the execution of the program.

Why do we use comments?

Comments help us to:

- Explain the purpose of code
- Make programs easier to understand
- Remember what a section of code does
- Make code easier to maintain

2. Types of Comments in C++

There are two common types of comments:

1. **Single-line comments**
2. **Multi-line comments**

3. Single-Line Comments

A **single-line comment** is used to write a comment on one line.

It starts with:

```
//
```

Everything after `//` on that line is treated as a comment.

Example 1

```
#include <iostream>
using namespace std;

int main() {
    // This is a single-line comment

    cout << "Hello World";

    return 0;
}
```

Output

```
Hello World
```

The compiler ignores:

```
// This is a single-line comment
```

Example 2: Comment After Code

```
int age = 20; // Store student's age
```

Here:

- `int age = 20;` → actual code
- `// Store student's age` → comment

4. Multi-Line Comments

A **multi-line comment** is used when the comment covers **more than one line**.

It starts with:

```
/*
```

and ends with:

```
*/
```

Example

```
#include <iostream>
using namespace std;

int main() {

    /*
       This program displays
       a simple message
       on the screen.
    */
}
```

```
    cout << "Hello World";  
  
    return 0;  
}
```

Output

Hello World

Everything between:

```
/*
```

and:

```
*/
```

is ignored by the compiler.

5. Difference Between Single-Line and Multi-Line Comments

Single-Line Comment	Multi-Line Comment
Used for one line	Used for multiple lines
Starts with //	Starts with /*
Ends at the end of the line	Ends with */
Example: // My comment	Example: /* My comment */

6. Complete Example

```
#include <iostream>  
using namespace std;  
  
int main() {  
  
    // Declare two numbers  
    int a = 10;  
    int b = 20;
```

```
/*  
    Add the two numbers  
    and display the result.  
*/  
int sum = a + b;  
  
cout << sum;  
  
return 0;  
}
```

Output

```
30
```

Quick Revision

Single-Line Comment

```
// This is a comment
```

Multi-Line Comment

```
/*  
    This is a  
    multi-line comment  
*/
```

Remember

Comments are written for programmers, not for the compiler.

They help make C++ programs **clear, readable, and easy to understand**.